

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split

from sklearn.metrics import confusion_matrix
from sklearn.metrics import recall_score
from sklearn.metrics import accuracy_score
from sklearn.metrics import precision_score
```

```
In [3]: df = pd.read_csv('Iris.csv')
```

```
In [4]: df.head()
```

```
Out[4]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [5]: df.describe
```

```
Out[5]: <bound method NDFrame.describe of
thCm PetalWidthCm \
0      1          5.1      3.5      1.4      0.2
1      2          4.9      3.0      1.4      0.2
2      3          4.7      3.2      1.3      0.2
3      4          4.6      3.1      1.5      0.2
4      5          5.0      3.6      1.4      0.2
..     ...      ...      ...      ...      ...
145    146        6.7      3.0      5.2      2.3
146    147        6.3      2.5      5.0      1.9
147    148        6.5      3.0      5.2      2.0
148    149        6.2      3.4      5.4      2.3
149    150        5.9      3.0      5.1      1.8

Species
0      Iris-setosa
1      Iris-setosa
2      Iris-setosa
3      Iris-setosa
4      Iris-setosa
..     ...
145    Iris-virginica
146    Iris-virginica
147    Iris-virginica
148    Iris-virginica
149    Iris-virginica

[150 rows x 6 columns]>
```

```
In [6]: X = df.iloc[:,1:5]
```

```
In [7]: y = df['Species']
```

```
In [9]: X_train,X_test,y_train,y_test = train_test_split(
        X,y,test_size=0.25,random_state=42
    )
```

```
In [11]: model = GaussianNB()
```

```
In [12]: model.fit(X_train,y_train)
```

```
Out[12]: GaussianNB ⓘ ?
          ► Parameters
```

```
In [13]: y_pred = model.predict(X_test)
```

```
In [15]: cm = confusion_matrix(y_test,y_pred)
        print(cm)
```

```
[[15  0  0]
 [ 0 11  0]
 [ 0  0 12]]
```

```
In [16]: accuracy = accuracy_score(y_test,y_pred)
print(accuracy)
```

1.0

```
In [20]: precision = precision_score(y_test,y_pred,average='macro')
print(precision)
```

1.0

```
In [19]: error_rate = 1-accuracy
print(error_rate)
```

0.0

```
In [21]: recall = recall_score(y_test,y_pred,average='macro')
print(recall)
```

1.0

DSBDA Practical 6 — Data Analytics III

Naïve Bayes Classification on Iris Dataset

Dataset Source: [Iris Dataset Kaggle](#)

Aim

To implement Naïve Bayes classification algorithm on Iris dataset and compute:

- Confusion Matrix
 - TP, TN, FP, FN
 - Accuracy
 - Error Rate
 - Precision
 - Recall
-

What is Naïve Bayes?

Naïve Bayes is a supervised machine learning classification algorithm based on:

- Bayes Theorem
- Probability

Used for:

- Email spam detection
 - Text classification
 - Medical diagnosis
-

Why Called “Naïve”?

Because it assumes:

- All features are independent

Even though in real life they may not be independent.

Bayes Theorem

Formula:

genui☆
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Where:

- $P(A|B)$ = Posterior probability
 - $P(B|A)$ = Likelihood
 - $P(A)$ = Prior probability
 - $P(B)$ = Evidence
-

About Iris Dataset

Contains flower measurements:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Classes:

- Iris-setosa
- Iris-versicolor

- Iris-virginica
-

Step 1: Import Libraries

Code

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
```

Explanation

Library	Purpose
pandas	Data handling
numpy	Numerical operations
train_test_split	Train-test splitting
GaussianNB	Naïve Bayes classifier
metrics	Evaluation

Viva Questions

Why GaussianNB used?

Because Iris dataset contains continuous numerical features.

Step 2: Load Dataset

Code

```
df = pd.read_csv("Iris.csv")
```

Display Dataset

Code

```
df.head()
```

Step 3: Check Dataset Information

Code

```
df.info()
```

Explanation

Shows:

- Datatypes
 - Rows and columns
 - Missing values
-

Step 4: Select Features and Target

Features

Code

```
X = df.iloc[:, 1:5]
```

Target Variable

Code

```
y = df['Species']
```

Explanation

```
iloc[:,1:5]
```

Selects:

- SepalLengthCm
 - SepalWidthCm
 - PetalLengthCm
 - PetalWidthCm
-

Viva Questions

Why not use Id column?

Id does not help prediction.

Step 5: Split Dataset

Code

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.25, random_state=0  
)
```

Explanation

Parameter	Meaning
test_size=0.25	25% testing
random_state=0	Same split every run

Step 6: Create Naïve Bayes Model

Code

```
model = GaussianNB()
```

Explanation

Creates Gaussian Naïve Bayes classifier.

Why “Gaussian”?

Assumes features follow:

- Normal distribution
-

Gaussian Distribution Formula

genui☆
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$
↕↗

Step 7: Train Model

Code

```
model.fit(X_train, y_train)
```

Explanation

Model learns probability patterns from training data.

Step 8: Predict Output

Code


```
y_pred = model.predict(X_test)
```

Explanation

Predicts flower species.

Step 9: Create Confusion Matrix

Code

```
cm = confusion_matrix(y_test, y_pred)

print(cm)
```

Example Output

```
[[13  0  0]
 [ 0 16  0]
 [ 0  0  9]]
```

Important Point

Iris dataset has:

3 classes

So confusion matrix becomes:

3 × 3

Meaning of Rows and Columns

Row	Actual class
Column	Predicted class

For Binary Classification Only

TP, TN, FP, FN are directly used mainly for:

2-class problems

But for practical exams, teachers usually calculate them using one class vs rest approach.

Example for Iris-setosa

Treat:

Iris-setosa = Positive

Others = Negative

Extract Values

Suppose confusion matrix:

```
[[13 0 0]
 [0 16 0]
 [0 0 9]]
```

For Setosa:

TP = 13

FP = 0

FN = 0

TN = 25

Because:

$16 + 9 = 25$

Step 10: Accuracy

Code

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy =", accuracy)
```

Accuracy Formula

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Step 11: Error Rate

Code

```
error_rate = 1 - accuracy  
  
print("Error Rate =", error_rate)
```

Step 12: Precision

Code

```
precision = precision_score(  
    y_test,  
    y_pred,  
    average='macro'  
)  
  
print("Precision =", precision)
```

Why `average='macro'` ?

Because Iris dataset has:

multiple classes

Macro average:

- Computes average precision for all classes.
-

Precision Formula

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Step 13: Recall

Code

```
recall = recall_score(  
    y_test,  
    y_pred,  
    average='macro'  
)  
  
print("Recall =", recall)
```

Recall Formula

$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$

Important Theory

What is Classification?

Predicting categories/classes.

Examples:

- Spam/Not Spam
 - Disease/No Disease
-

Types of Naïve Bayes

Type	Used For
GaussianNB	Continuous data
MultinomialNB	Text/count data
BernoulliNB	Binary data

Advantages of Naïve Bayes

- Simple

- Fast
 - Works well on small datasets
 - Good for text classification
-

Disadvantages

- Assumes feature independence
 - May not perform well if features highly correlated
-

What is Prior Probability?

Probability before observing data.

What is Posterior Probability?

Probability after observing data.

Important Viva Questions

Basic Questions

What is Naïve Bayes?

Probability-based classification algorithm.

Why called naïve?

Assumes features are independent.

What theorem is used?

Bayes theorem.

Why GaussianNB used?

Features are continuous.

Intermediate Questions

Difference between GaussianNB and MultinomialNB?

GaussianNB	MultinomialNB
Continuous data	Count/text data

What is confusion matrix?

Performance evaluation table.

What does diagonal of confusion matrix represent?

Correct predictions.

Advanced Questions

Why average='macro' used?

For multiclass evaluation.

Can Naïve Bayes work on correlated features?

Yes, but performance may reduce.

Why Naïve Bayes is fast?

Uses probability calculations and independence assumption.

External May Ask Modifications

Predict Single Sample

```
model.predict([[5.1,3.5,1.4,0.2]])
```

Display Probability

```
model.predict_proba(X_test)
```

Find F1 Score

```
from sklearn.metrics import f1_score  
  
print(f1_score(y_test, y_pred, average='macro'))
```

F1 Score Formula

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Common Exam Errors

Error	Reason
Wrong slicing	Incorrect feature selection
Missing average='macro'	Multiclass error
Using string instead of numeric	Wrong input type

2-Minute Viva Explanation

"In this practical, we implemented the Naïve Bayes classification algorithm on the Iris dataset. We imported required libraries, loaded the dataset, selected features and target variable, and split the dataset into training and testing sets. We used GaussianNB because the dataset contains continuous numerical features. The model was trained using fit() and predictions were generated using predict(). We evaluated the model using confusion matrix, accuracy, precision, recall, TP, TN, FP, and FN. Naïve Bayes is a fast probabilistic classification algorithm based on Bayes theorem and feature independence assumption."

This is one of the MOST COMMON DSBDA viva questions. External mainly wants to see whether you understand multiclass confusion matrix.

Important Concept

For:

3 × 3 confusion matrix

TP, TN, FP, FN are calculated:

“One class vs Rest”

Meaning:

- Take one class as Positive
 - Treat all other classes as Negative
-

Example 3×3 Confusion Matrix

Suppose output is:

```
[[13  0  0]
 [ 0 16  2]
 [ 1  0  8]]
```

Classes:

1. Iris-setosa
 2. Iris-versicolor
 3. Iris-virginica
-

Matrix Meaning

Actual \ Predicted	Setosa	Versicolor	Virginica
Setosa	13	0	0
Versicolor	0	16	2
Virginica	1	0	8

Step-by-Step Calculation for ONE Class

Case 1: Taking Iris-setosa as Positive Class

Treat:

Setosa = Positive

Others = Negative

TP (True Positive)

Correctly predicted Setosa.

From matrix:

13

So:

TP = 13

FN (False Negative)

Actual Setosa but predicted something else.

Look at Setosa row excluding TP:

0 + 0 = 0

So:

FN = 0

FP (False Positive)

Predicted Setosa but actually other class.

Look at Setosa column excluding TP:

0 + 1 = 1

So:

FP = 1

TN (True Negative)

Everything else.

Formula:

TN = Total - TP - FP - FN

Total elements:

$13+0+0+0+16+2+1+0+8 = 40$

Therefore:

$TN = 40 - 13 - 1 - 0$

$TN = 26$

Final Values for Setosa

Metric	Value
TP	13
FP	1
FN	0
TN	26

Easiest Way to Explain to External

Say this EXACT sentence:

"In multiclass confusion matrix, TP, FP, FN, and TN are calculated using one-vs-rest approach. One class is treated as positive and all remaining classes are treated as negative."

This line alone impresses externals.

Super Important Visual Trick

For any class:

TP

Diagonal element of that class.

Example:

```
Setosa TP = cm[0][0]  
Versicolor TP = cm[1][1]  
Virginica TP = cm[2][2]
```

FN

Same row excluding diagonal.

Because:

- Actual class same
 - Prediction wrong
-

FP

Same column excluding diagonal.

Because:

- Prediction same
 - Actual class wrong
-

TN

Everything remaining.

Example for Versicolor

Confusion matrix again:

```
[[13  0  0]
 [ 0 16  2]
 [ 1  0  8]]
```

TP

16

FN

Same row excluding TP:

$0 + 2 = 2$

FP

Same column excluding TP:

$0 + 0 = 0$

TN

$40 - 16 - 2 - 0 = 22$

Final Values

Metric	Value
TP	16
FP	0
FN	2
TN	22

Common Viva Questions

Q. Can TP/TN/FP/FN be directly calculated in multiclass classification?

Not directly like binary classification.

We use:

One-vs-Rest approach

Q. Why diagonal elements important?

They represent:

Correct predictions

Q. What do off-diagonal elements represent?

Misclassifications.

Q. How to calculate FP?

Take:

- Same column
 - Excluding diagonal
-

Q. How to calculate FN?

Take:

- Same row
 - Excluding diagonal
-

VERY IMPORTANT MEMORY TRICK

For any selected class:

Metric	How to Find
TP	Diagonal element
FN	Row sum excluding TP

Metric	How to Find
FP	Column sum excluding TP
TN	Remaining all elements

Short Viva Answer (Best)

"For multiclass confusion matrix, TP, TN, FP, and FN are calculated using one-vs-rest method. For a selected class, the diagonal element is TP, remaining values in same row are FN, remaining values in same column are FP, and all remaining elements are TN."